

ARSHIA AKHAVAN

Master's Student in Computer Science

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EXPERIENCE

Software Engineer

Sotoon

📅 Jan 2023 – Dec 2024

📍 Tehran, Iran

Sotoon is a B2B cloud provider that focuses on different aspects of Cloud Computing, such as hardware abstraction, providing storage servers, DNS, CDN, Bigdata solutions, and managed k8s clusters. I worked in the K8s as a Service team. Experiences:

- Using **SIGS/Cluster-api** to provide bootstrapping and operations as a service for k8s.
- Develop and Maintain SotoonCSI, a **CSI implementation** that integrates with Sotoon's storage backend.
- Develop **node-AutoScaler** for k8s cluster, which integrates with Sotoon's Computation backend to reduce customers' costs.
- Develop and maintain a **CCM** module to provide load balancers for k8s clusters using Sotoon's compute infrastructure.
- Provide additional cluster plugins such as **DNS** and **CNI** as an automated solution.
- Plan and Lead **Migration** of **over 100 customers'** k8s-clusters to the new infrastructure with **zero downtime**.

Data Platform Engineer

Digikala

📅 Aug 2021 – Jan 2023

📍 Tehran, Iran

Digikala is the largest e-commerce company in Iran with 40+ million users. As a member of the Bigdata & AI team, my job is to provide the necessary infrastructure needed to store and process the data accumulated from the users.

Experiences:

- Developing **Karavan**, a **Parallel Processing Engine** for data pipelines on top of **Apache Spark**.
- **Reducing Computation time** of our Pipelines by **revising our data retrieval models and Queries**.
- **Reducing latency** of our API by locating **critical paths** in the codebase of the framework and **optimizing** them.
- Using **K8s** as the main orchestrator with **Ceph** as our storage backend; provided a variety of tools for the DS/DE team to work with, including **Airflow** as our job scheduler and **Spark** as our main batch/stream processor and pipeline engine.
- maintain **Kafka** cluster as a Sync and **Debezium** to sync external Databases to it.
- Manage the platform's infrastructure based on **GitOps** protocols using **argoCD**. Automate deployment of our APIs using ArgoCD with **Kustomize** and **Helm charts**.
- Deploy and maintain a staging environment and build a framework for our Data Science team to better interact with our infrastructure.

DevOps Engineer

Tapsell

📅 Nov 2019 – Aug 2021

📍 Tehran, Iran

Providing the infrastructure needed by other technical teams in the corporation.

Experiences:

- Managing and automating deployment procedures of databases such as **MongoDB**, **Cassandra**, and **ElasticSearch** with **Ansible**
- Optimizing Cassandra's **Disk usage** for different workloads
- Developing a fully automated backup, validity check, and restore tool for managing our backups in a cloud native infrastructure.
- Deploy and maintain a fully functional staging cluster using **k8s** as its main orchestrator.
- Develop CI/CD flows for our company using **GitlabCI**.
- Working with **Terraform** to automate deploying our infrastructure on **VMWare**.
- Administering Linux-based virtual machines, addressing network issues and bottlenecks, and improving our overall performance based on it.

PUBLICATIONS

Journal Articles

- A. Akhavan, A. Hosseinpour, A. Heydarnoori, and M. Keshani, "Linkanchor: An autonomous llm-based agent for issue-to-commit link recovery," *arXiv preprint*, vol. arXiv:2508.12232, 2025, Submitted 17 August 2025; revised 2 September 2025. DOI: [10.48550/arXiv.2508.12232](https://doi.org/10.48550/arXiv.2508.12232).

VOLUNTEER EXPERIENCE

Technical Chapter

Student's Scientific Chapter (SSC)

Aug 2020 – Aug 2021 Tehran, Iran

Maintain Technical resources owned by SSC and supervise technical requirements of events held by SSC

Chief of Infrastructure and System Team

Sharif AI Challenge 2021

Dec 2020 – May 2021 Tehran, Iran

Develop and deploy a scalable infrastructure/software for AI challenge's high demanding game engine. [GitHub](#)

Chief Technical Staff

ICPC

May 2023 Tehran, Iran

Build contest-customized OS images and distribute them on contestant hosts using Network boot.
Manage and configure hosts used by contestants and monitor their connectivity.
Restrict external access and host-to-host access of each contestant's host device.

Chief Technical Staff

ICPC west championship

Dec 2019 Tehran, Iran

Manage and configure hosts used by contestants and monitor their connectivity

EDUCATION

M.Sc. in Computer Science

San Diego State University

Sep 2025 – Currently CA, USA

B.Sc. in Computer Engineering

Sharif University of Technology

Sep 2018 – Apr 2024 Iran

Mathematics and Physics Diploma

Helli High-school (Sampad)

Sep 2010 – Jul 2018 Iran

HONORS & AWARDS



BGSU ACM Hackathon

Awarded second place for designing [pigniq](#)

PROGRAMMING LANGUAGES

Rust

Go

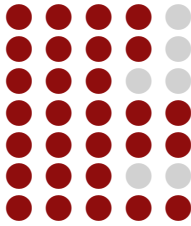
C

Bash

Python

Java

SQL



TECHNOLOGIES

AVX Cuda OpenMP rayon-rs crossbeam-rs

k8s Docker Docker-swarm Nginx HAproxy

Ansible terraform GitlabCI ArgoCD

Spark Airflow

Ceph minIO Kafka Redis Cassandra MongoDB

LANGUAGES

English

French

Persian